Dual-Server Independent Domination in Graphs

Mustapha Chellali[1], Randy R. Davila[2,3,*], Teresa W. Haynes[4,5],
Stephen T. Hedetniemi[6], and Ryan Pepper[7]

[1]LAMDA-RO Laboratory, Department of Mathematics, University of Blida, B.P. 270, Blida, Algeria
[2]First Principles, 100 University Ave, Toronto, ON M5J 1V6, Canada
[3]Department of Computational Applied Mathematics & Operations Research, Rice University, 6100 Main Street, Houston, TX 77005, USA
[4]Department of Mathematics and Statistics, East Tennessee State University, Johnson City, TN 37614, USA
[5]Department of Mathematics and Applied Mathematics, University of Johannesburg, Auckland Park, 2006, South Africa
[6]Emeritus College, Clemson University, Clemson, SC 29634, USA
[7]Department of Mathematics and Statistics, University of Houston–Downtown, Houston, TX 77002, USA
[*]Corresponding author: `randy@firstprinciples.org`

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Abstract

For a graph $G = (V, E)$, a set $S \subseteq V(G)$ is a *dual-server dominating set* if S can be partitioned into two subsets R and B such that every vertex in $V(G) \setminus S$ is adjacent to at least one vertex in R and at least one vertex in B. In this paper, we introduce and study two independent versions of *dual-server domination.* Specifically, if S itself is independent, then S is an *independent dual-server dominating set*; while if R and B are independent sets, with no additional independence requirement on S, then S is a *dual-server independent dominating set.* The latter notion is the primary focus of the paper. We prove that dual-server independent domination is well-defined for every nontrivial graph, establish basic comparison bounds, determine exact values for several standard graph families, show that the parameter agrees with both *dual-server domination* and *2-domination* on trees, and derive sharp bounds in terms of *order*, *isolation number*, and *diameter*, including a characterization of equality for the isolation bound. We also give a path-covering and matching upper bound for dual-server domination, and hence, for DSI-domination on trees. Finally, we show that, although the associated decision problem is NP-complete even for chordal bipartite graphs, the parameter admits a compact binary integer programming formulation and is polynomial-time solvable on graph classes of bounded *treewidth.* The work also records several roles played by the AI-assisted system `Theo-Conjecture`: its *Add Definition* tool led to the binary integer programming formulation, its conjecture searches suggested several of the bounds proved here, and its surviving conjectures, after human and *Co-Pilot* counterexample attempts, form the concluding open-problems section.

Keywords: Domination; independent domination; dual-server domination; 2-domination; isolation number; path cover; matching number; diameter; treewidth; NP-completeness.

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1 Introduction

Domination in graphs has been extensively studied for more than fifty years, with numerous variations developed to model different structural and applied constraints. For a comprehensive treatment of domination in graphs, we refer the reader to the books [22, 23, 24]. A set $S \subseteq V(G)$ is a *dominating set* of a graph G if every vertex in $V(G) \setminus S$ has a neighbor in S, and the minimum cardinality of such a set is the *domination number* $\gamma(G)$. A dominating set S is an *independent dominating set* of G if S is an independent set in G, and the minimum cardinality of such a set is the *independent domination number* $i(G)$. Many variants of domination strengthen this requirement by asking that vertices outside the dominating set receive several forms of coverage, or by imposing internal structure on the dominating set itself. The *independent domination number*, the *2-domination number*, and related multiple-domination parameters are examples of this general pattern.

Dual-server domination was introduced recently in [7] as a two-service version of this domination. A set $S \subseteq V(G)$ is a dual-server dominating set if S admits a partition $S = R \cup B$ such that every vertex in $V(G) \setminus S$ has a neighbor in R and a neighbor in B. Thus, vertices outside S are served by both classes, which may be interpreted as two distinct types of service. The partition is part of the structure: the same set S may admit more than one dual-server partition, and the choice of partition may affect additional restrictions placed on the two classes.

In this paper, we study independent restrictions on dual-server domination. There are two natural possibilities. The more restrictive one requires S itself to be independent, giving the direct analogue of independent domination. The second requires only the two classes R and B to be independent, allowing edges between the two classes but forbidding edges within either class. We call these notions *independent dual-server domination* and *dual-server independent domination*, respectively, and abbreviate them as IDS-domination and DSI-domination. Accordingly, feasible sets for these two notions will be called IDS-dominating sets and DSI-dominating sets. The latter notion is the principal object of study here. It retains the two-service interpretation of dual-server domination while imposing independence on each service class, and, unlike the former, it exists for every nontrivial graph.

Although our focus is graph-theoretic, the two-color viewpoint also suggests a modeling interpretation. In ensemble learning, diversity among predictors is a central design theme [8, 14, 30], and negative correlation learning explicitly penalizes correlated behavior [31]. If vertices represent learners or data sources and adjacency records a strong dependence relation, then a DSI-dominating set selects two internally diverse representative panels that jointly cover the remaining vertices. A similar two-panel interpretation may also be relevant to multi-agent AI systems, where autonomous agents interact, cooperate, and may require oversight or debate mechanisms [15, 29, 34, 36]. Appendix A records two baseline computational illustrations of this interpretation; these are included only as motivation and are not part of the main mathematical development of the paper.

`Theo-Conjecture` is an AI-assisted automated conjecturing system implemented within First Principles and developed in the lineage of TxGraffiti and Graffiti3 [12, 13]. At a high level, the system maintains databases of mathematical objects and computable invariants, searches those data for candidate relationships, tests proposed statements against available examples, and exports selected conjectures for human study. `Theo-Conjecture` played three complementary roles in the development of this work. First, its *Add Definition* tool was used to introduce γ_{dsi} as a new graph invariant in the First Principles workspace database. In that workflow, a researcher supplies a mathematical description of an invariant, the system drafts an executable computation, and the draft is checked on small examples before being used in larger searches. For γ_{dsi}, this process led to an exact binary integer programming model. We include the mathematical form of that model in Section 6, both to make the computational experiments reproducible and to make explicit how the two-color definition was encoded.

Second, after γ_{dsi} was available as a computable invariant, `Theo-Conjecture` was used to search for candidate inequalities involving standard graph parameters and structural graph classes. In particular, the isolation lower bound in Theorem 20, the diameter lower bound in Theorem 24, and the decycling-number specialization in Proposition 28 arose from this process.

Third, the open-problems section presents a curated list of remaining conjectures generated by `Theo-Conjecture` that survived both human counterexample searches and counterexample attempts through the *Co-Pilot* chat interface. In this sense, the automated system served as a source of definitions, computations, examples, comparisons, and problems, whereas the statements, proofs, refinements, and interpretations that follow are the mathematical contributions of the authors.

The paper is organized as follows. We prove the existence of dual-server independent domination for every nontrivial graph and establish comparison chains with other domination parameters. We determine values of dual-server domination parameters for several graph families and show that on every nontrivial tree, DSI-domination agrees with both dual-server domination and 2-domination. We then prove sharp bounds involving order, isolation number, and diameter, determine a path-covering and matching bound that is especially useful on trees, and give hereditary induced-subgraph bounds involving forest and bipartite numbers. Additionally, we give a compact binary integer programming formulation, prove NP-completeness even for chordal bipartite graphs, and derive a tractability result for bounded-treewidth instances. We conclude with an open-problems section featuring selected conjectures generated using `Theo-Conjecture`.

All graphs in this paper are finite and simple. Let $G = (V, E)$ be a graph with vertex set $V = V(G)$, edge set $E = E(G)$, and *order* $n = n(G) = |V(G)|$. Let $\overline{G}$ denote the *complement* of G. For a set $S \subseteq V(G)$, we denote the subgraph induced by S by $G[S]$. For an integer $k \geq 1$, we use the standard notation $i \in [k]$ to mean that i is an integer and $1 \leq i \leq k$. The *open neighborhood* of a vertex v in G is $N(v) = \{u \in V(G) : uv \in E\}$ and the *closed neighborhood of* v is $N[v] = \{v\} \cup N(v)$. The *degree* of a vertex v, denoted by $\deg(v)$, is the cardinality of its open neighborhood. The *open neighborhood of a set* $S \subseteq V(G)$ is $N(S) = \bigcup_{v \in S} N(v)$, while the *closed neighborhood of a set* $S \subseteq V(G)$ is the set $N[S] = \bigcup_{v \in S} N[v]$. The *minimum degree* and *maximum degree* of G are $\delta(G)$ and $\Delta(G)$, respectively. A vertex of degree one is a leaf, and its unique neighbor is a support vertex. The maximum cardinality of an independent set of G is the *independence number* $\alpha(G)$. A graph is *nontrivial* if it has at least two vertices. We write P_n, C_n, and K_n for the path, cycle, and complete graph of order n, respectively, and $K_{r,s}$ for the complete bipartite graph with partite sets of sizes r and s, where $1 \leq r \leq s$.

2 Definitions and Preliminary Results

We first formally define the three dual-server domination concepts that will be compared throughout the paper. The terminology is intentionally close, so we emphasize the order of the words: in an *independent dual-server* set the whole selected set is independent, whereas in a *dual-server independent* set the two server classes are independent separately.

Definition 1 (Dual-Server Dominating Set)**.** Let G be a graph. A set $S \subseteq V(G)$ is a *dual-server dominating set*, or *DS-dominating set*, if there is a partition $S = R \cup B$ into two nonempty sets such that every vertex in $V(G) \setminus S$ has at least one neighbor in R and at least one neighbor in B. Such a partition is called a *DS-partition* of S. The minimum cardinality of a DS-dominating set of G is the *dual-server domination number*, denoted $\gamma_{\rm ds}(G)$. A DS-dominating set of cardinality $\gamma_{\rm ds}(G)$ is called a $\gamma_{\rm ds}$-set of G.

Definition 2 (Independent Dual-Server Dominating Set)**.** An *independent dual-server dominating set*, or *IDS-dominating set*, of G is a DS-dominating set S such that S is an independent set of G. If such a set exists, then the minimum cardinality of an IDS-dominating set is denoted $i_{\rm ds}(G)$. An IDS-dominating set of cardinality $i_{\rm ds}(G)$ is called an $i_{\rm ds}$-set of G.

Definition 3 (Dual-Server Independent Dominating Set)**.** A *dual-server independent dominating set*, or *DSI-dominating set*, of G is a DS-dominating set S with an associated partition $S = R \cup B$ such that both R and B are independent sets of G. Such a partition is called a *DSI-partition* of S. The minimum cardinality of a DSI-dominating set is the *dual-server independent domination number*, denoted $\gamma_{\rm dsi}(G)$. A DSI-dominating set of cardinality $\gamma_{\rm dsi}(G)$ is called a $\gamma_{\rm dsi}$-set of G.

Since every vertex outside a dual-server dominating set must have at least two neighbors in the set, we note the following.

Observation 1. *Every vertex of degree at most one belongs to every DS-dominating set, every DSI-dominating set, and every IDS-dominating set of G.*

Throughout, whenever a particular partition $S = R \cup B$ is fixed, we refer to R and B as the red and blue color classes, respectively, and to the partition as a *dual-server coloring.* Thus, we form such a partition by coloring the vertices of S red and blue and the vertices in $V(G) \setminus S$ white. We also use the term selected for vertices in the set S, and unselected for vertices in $V(G) \setminus S$. We say that an edge is a red-blue edge if it is incident to both a red vertex and a blue vertex.

Every IDS-dominating set is a DSI-dominating, but not conversely, since DSI allows red-blue edges in $G[S]$. This distinction is significant, making DSI-domination the more robust notion as we next show that it exists for all nontrivial graphs, whereas IDS-domination does not. For example, the nontrivial complete graph K_n does not have an IDS-set, but $\gamma_{\rm dsi}(K_n) = 2$. We now show that the parameter $\gamma_{\rm dsi}(G)$ is well-defined for every nontrivial graph.

Theorem 2. *Every nontrivial graph has a DSI-dominating set.*

Proof. Let G be a graph of order $n \geq 2$, and let I be a minimum independent dominating set of G. If $I = V(G)$, then $G = \overline{K}_n$. Since $n \geq 2$, color one vertex of I red and all remaining vertices blue. The resulting partition is vacuously a DSI-partition.

Assume now that $V(G) \setminus I \neq \emptyset$. Let J be a minimum independent dominating set of the induced subgraph $G[V(G) \setminus I]$. Set $S = I \cup J$, color the vertices of I red, and color the vertices of J blue. Both color classes are independent. Moreover, each vertex in $V(G) \setminus S$ lies outside I, and hence, has a red neighbor in I; it also lies outside J in $G[V(G) \setminus I]$, and hence has a blue neighbor in J. Therefore, S is a DSI-dominating set of G. □

Recall that $i(G)$ and $\alpha(G)$ denote the independent domination number and the independence number of G, respectively.

Corollary 3. *If G is a graph of order $n \geq 2$, then*

$$\gamma_{\rm dsi}(G) \leq \min\{n,\, i(G) + \alpha(G)\}.$$

Proof. Let I be a minimum independent dominating set of G. If $I = V(G)$, then the proof of Theorem 2 gives a DSI-dominating set of cardinality $n = i(G)$. Otherwise, the same proof chooses an independent dominating set J of $G[V(G) \setminus I]$, and $|J| \le \alpha(G)$. Hence, $\gamma_{\rm dsi}(G) \le |I| + |J| \le i(G) + \alpha(G)$. □

The following result gives a sharp bound in terms of order.

Theorem 4. *If G is a connected graph of order $n \ge 3$, then*

$$\gamma_{\rm dsi}(G) \le n - 1.$$

Moreover, the bound is sharp for every $n \ge 3$.

Proof. Since G is connected and $n \ge 3$, there is a vertex v with degree at least two. Choose two distinct neighbors x and y of v. Among all pairs (R, B) of disjoint subsets of $V(G) \setminus \{v\}$ such that $x \in R$, $y \in B$, and both R and B are independent, choose one that is maximal with respect to inclusion. Such a pair exists, since $(\{x\}, \{y\})$ is admissible.

Let $S = R \cup B$. The vertex v has a neighbor in each of R and B, namely x and y, respectively. Now let $w \in V(G) \setminus (S \cup \{v\})$. If w has no neighbor in R, then $R \cup \{w\}$ is independent, contradicting the maximality of R. Thus, w has a neighbor in R. The same argument, with the colors interchanged, shows that w has a neighbor in B. Hence, S, with partition $S = R \cup B$, is a DSI-dominating set of G. Since $v \notin S$, we have $|S| \le n - 1$, and therefore, $\gamma_{\rm dsi}(G) \le n - 1$.

The bound is sharp for the star $K_{1,n-1}$ where $n \ge 3$. By Observation 1, every leaf belongs to every DSI-dominating set, and hence, $\gamma_{\rm dsi}(K_{1,n-1}) \ge n - 1$. Conversely, selecting all leaves and coloring one leaf red and the remaining leaves blue gives a DSI-dominating set. Therefore, $\gamma_{\rm dsi}(K_{1,n-1}) = n - 1$. □

Corollary 5. *If G is a nontrivial claw-free graph, then*

$$\gamma_{\rm dsi}(G) \le 3\gamma(G).$$

Proof. Allan and Laskar [2] proved that $i(G) = \gamma(G)$ for claw-free graphs, and Sumner [35] proved that $\alpha(G) \le 2\gamma(G)$ for claw-free graphs. The result follows from Corollary 3. □

If we restrict ourselves to claw-free graphs of degree at least 3, we know that $\gamma(G) \le \frac{n}{3}$ (see [24]). Thus, Corollary 5 gives the trivial bound $3\gamma(G) \le n$ for this class of graphs. The next result provides an improvement on this bound reducing it to $2n/3$. Note that bound $2n/3$ is reached for the Cartesian product $K_3 \Box K_2$.

Proposition 6. *If G is a nontrivial claw-free graph of order n and minimum degree $\delta \ge 3$,*

$$\gamma_{\rm dsi}(G) \le \frac{n + \gamma(G)}{2}.$$

Proof. Let I be a minimum independent dominating of G, and note that $|I| = i(G) = \gamma(G)$. Every vertex in $V(G) \setminus I$ has at least one neighbor in I but no more than two such neighbors, otherwise a claw is formed. Since $\delta \ge 3$, every vertex in $V(G) \setminus I$ has at least one neighbor in $V(G) \setminus I$, and thus the subgraph induced by $V(G) \setminus I$ is isolate-free. Let J be a minimum independent dominating set of $G[V(G) \setminus I]$, and note that $|J| \le \frac{|V(G) \setminus I|}{2}$. It now follows that every vertex in

$V(G) \setminus (I \cup J)$ has a neighbor in I and another one in J. If, in addition, the vertices of I are colored red and those of J are colored blue, then $I \cup J$ is a DSI-dominating set of G. Hence, $\gamma_{\rm dsi}(G) \le |I \cup J| \le |I| + \frac{|V(G)\setminus I|}{2} = \frac{n+\gamma(G)}{2}$. $\square$

We also present the following inequalities involving dual-server domination and other domination parameters. The *2-domination number*, denoted $\gamma_2(G)$, is the minimum cardinality of a set $D \subseteq V(G)$ such that every vertex in $V(G) \setminus D$ has at least two neighbors in D.

Observation 7. *For every nontrivial graph G,*

$$\gamma(G) \le \gamma_2(G) \le \gamma_{\rm ds}(G) \le \gamma_{\rm dsi}(G).$$

Observation 8. *If G has an IDS-dominating set, then*

$$\gamma(G) \le i(G) \le i_{\rm ds}(G) \le \alpha(G).$$

and

$$\gamma_{\rm dsi}(G) \le i_{\rm ds}(G) \le \alpha(G).$$

In a *well-covered graph*, every maximal independent set is maximum, and hence, $i(G) = \alpha(G)$. The next result follows from Observation 8.

Observation 9. *If G is well-covered and has an IDS-dominating set, then*

$$i(G) = i_{\rm ds}(G) = \alpha(G).$$

Blidia, Chellali, and Favaron [3] proved that $\gamma_2(G) \ge \alpha(G)$ for claw-free graphs. Blidia, Chellali, and Volkmann [5] proved the same inequality for block graphs and for graphs having at most one cycle. The next chain follows from this inequality together with the inequalities in Observations 7 and 8.

Observation 10. *Let G be a graph that has an IDS-dominating set. If G is claw-free, or if G is a block graph, or if G has at most one cycle, then*

$$\gamma_2(G) = \gamma_{\rm ds}(G) = \gamma_{\rm dsi}(G) = i_{\rm ds}(G) = \alpha(G).$$

These parameters are also close in spirit to *H-forming domination*, as introduced by Haynes, Hedetniemi, Henning, and Slater [25]. In the case $H = P_3$, a *P_3-forming dominating set S* of a graph G requires that each vertex in $V(G) \setminus S$ form a copy of a path P_3 with two vertices in S. The minimum cardinality of any P_3-forming dominating set is the *P_3-forming domination number*, denoted $\gamma_{P_3}(G)$. If a P_3-forming dominating set is independent, then it is also an *independent 2-dominating set*, whose minimum cardinality is denoted by $i_2(G)$. As with IDS-domination, such sets need not exist; the cycle C_5 is a standard example. Whenever both $i_2(G)$ and $i_{\rm ds}(G)$ exist, every IDS-dominating set is an independent 2-dominating set, and hence,

$$\gamma_{P_3}(G) \le i_2(G) \le i_{\rm ds}(G).$$

3 Examples and Exact Values

We next determine the DSI-domination number for several familiar graph classes. These examples are included not merely as checks on the definitions, but also to show how the two independent parameters vary. Complete graphs show that DSI-domination may exist in its smallest possible form even when IDS-domination does not exist at all, while paths and cycles show the parity constraints that arise when the dual-server dominating set is required to be independent. We begin by stating the values for complete graphs as mentioned in Section 2.

Observation 11. *For every $n \geq 2$,*

$$\gamma_{\mathrm{ds}}(K_n) = \gamma_{\mathrm{dsi}}(K_n) = 2.$$

Moreover, K_n has no IDS-dominating set for $n \geq 2$.

Next we consider complete bipartite graphs $K_{r,s}$.

Proposition 12. *Let $1 \leq r \leq s$. If $r = s = 1$, then $\gamma_{\mathrm{dsi}}(K_{r,s}) = 2$ and $K_{r,s}$ does not have an IDS-dominating set. Otherwise,*

$$\gamma_{\mathrm{dsi}}(K_{r,s}) = i_{ds}(K_{r,s}) = \begin{cases} s, & r = 1 < s, \\ r, & r \geq 2. \end{cases}$$

Proof. Let X and Y be the partite sets, with $|X| = r \leq s = |Y|$. If $r = s = 1$, that is, $K_{r,s} = K_2$, then $\gamma_{\mathrm{dsi}}(K_2) = 2$ and K_2 does not have an IDS-dominating set.

If $r = 1 < s$, then every vertex in Y has degree one and must belong to every DSI-dominating set by Observation 1. Thus, $s \leq \gamma_{dsi}(K_{r,s})$. Conversely, selecting all vertices of Y, with at least one red vertex and at least one blue vertex, gives an IDS-dominating set: the unique vertex in X is the only unselected vertex and it sees both colors. Hence, $\gamma_{\mathrm{dsi}}(K_{1,s}) \leq i_{\mathrm{ds}}(K_{1,s}) \leq s$, and so, $\gamma_{\mathrm{dsi}}(K_{1,s}) = i_{\mathrm{ds}}(K_{1,s}) = s$ for $s \geq 2$.

Assume $r \geq 2$. Selecting all vertices of X, with at least one red and one blue vertex, gives an IDS-dominating set of cardinality r. Hence, $\gamma_{\mathrm{dsi}}(K_{r,s}) \leq i_{\mathrm{ds}}(K_{r,s}) \leq r$. Conversely, if a DSI-dominating set omits a vertex from both partite sets, then the unselected vertex in X requires both colors in Y, and the unselected vertex in Y requires both colors in X. This would place adjacent vertices of the same color in the complete bipartite graph, contrary to the independence of each color class. Thus, any DSI-dominating set contains one entire partite set, and therefore, has cardinality at least r. $\square$

Next we give values for paths and cycles.

Proposition 13. *For every $n \geq 2$,*

$$\gamma_{\mathrm{dsi}}(P_n) = \left\lceil \frac{n+1}{2} \right\rceil.$$

Moreover, P_n has an IDS-dominating set if and only if n is odd, and in that case

$$i_{\mathrm{ds}}(P_n) = \frac{n+1}{2}.$$

Proof. Write $P_n = v_1 v_2 \cdots v_n$. By Observation 1, the leaves v_1 and v_n belong to every DSI-dominating set. Moreover, every unselected internal vertex must have both of its path-neighbors selected, and those two neighbors must receive different colors. Hence, the unselected vertices form an independent subset of $\{v_2, \ldots, v_{n-1}\}$, of size at most $\lfloor (n-1)/2 \rfloor$. Thus, every DSI-dominating set has cardinality at least

$$n - \left\lfloor \frac{n-1}{2} \right\rfloor = \left\lceil \frac{n+1}{2} \right\rceil.$$

Equality is attained by leaving unselected the even-indexed internal vertices $v_2, v_4, \ldots$, and coloring the selected vertices alternately red and blue along the path. This proves the stated value of $\gamma_{\mathrm{dsi}}(P_n)$; it is the same value obtained for $\gamma_{\mathrm{ds}}(P_n)$ in [7].

For the IDS statement, the leaves of P_n must again belong to every IDS-dominating set. Since v_1 is selected and the selected set is independent, v_2 is unselected; then v_3 must be selected so that v_2 sees two selected neighbors. Continuing along the path forces the selected vertices to be precisely the vertices in odd positions. This set is an IDS-dominating set exactly when the last vertex also has odd position, that is, when n is odd. Its cardinality is then $(n+1)/2$, giving the asserted value of $i_{\mathrm{ds}}(P_n)$. □

Proposition 14. *For $n \geq 3$,*

$$\gamma_{\mathrm{dsi}}(C_n) = \begin{cases} \left\lceil \dfrac{n+2}{2} \right\rceil, & n \equiv 1, 2 \pmod 4, \\ \left\lceil \dfrac{n}{2} \right\rceil, & n \equiv 0, 3 \pmod 4. \end{cases}$$

Furthermore, C_n has an IDS-dominating set if and only if $n \equiv 0 \pmod 4$; in that case, $i_{\mathrm{ds}}(C_n) = n/2$.

Proof. Let $S = R \cup B$ be a DSI-dominating set of C_n, and let $W = V(C_n) \setminus S$. Every vertex of W has degree two, so its two neighbors must both belong to S, one in R and one in B. In particular, no two vertices of W are consecutive, and hence,

$$|W| \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Equivalently, $|S| \geq \lceil n/2 \rceil$.

We next determine when this lower bound on $|S|$ is attainable. Suppose first that $|W| = \lfloor n/2 \rfloor$. Then, in cyclic order, every two consecutive selected vertices either are adjacent in C_n or have exactly one white vertex between them. In either case, the two selected vertices must have different colors: adjacent selected vertices cannot lie in the same color class, and a white vertex must see both colors. Therefore, the selected vertices in cyclic order must be colored alternately. This is possible if and only if the number of selected vertices, namely $\lceil n/2 \rceil$, is even. Thus, $|S| = \lceil n/2 \rceil$ precisely when $n \equiv 0, 3 \pmod 4$.

If $n \equiv 1, 2 \pmod 4$, the preceding paragraph shows that one cannot have $|S| = \lceil n/2 \rceil$, so $|S| \geq \lceil n/2 \rceil + 1$. This bound is attained by taking $\lceil n/2 \rceil + 1$ selected vertices in cyclic order, coloring them alternately red and blue, and inserting the remaining $\lfloor n/2 \rfloor - 1$ vertices as isolated white vertices between selected pairs. Since the number of selected vertices is then even, the alternating coloring closes consistently around the cycle. These two cases give exactly the displayed formula for $\gamma_{\mathrm{dsi}}(C_n)$.

If S is required to be independent, then every unselected vertex still has both neighbors selected, while no two selected vertices are adjacent. Hence, selected and unselected vertices must alternate around the cycle. Thus, $|S| = n/2$ and n is even. The two colors of the selected vertices must also alternate around the cycle, which is possible exactly when $n/2$ is even, equivalently $n \equiv 0 \pmod 4$. □

4 Trees

The difference between the dual-server domination number and the dual-server independent domination number can be arbitrarily large. For example, consider the complete bipartite graph $G = K_{r,s}$, where $5 \le r \le s$. It is straightforward to check that $\gamma_{\mathrm{ds}}(G) = 4$, and by Proposition 12, we have $\gamma_{\mathrm{ds}}(G) = 4 < r = \gamma_{\mathrm{dsi}}(G)$. On the other hand, we next show that the two parameters are equal for trees. Our proof is a recoloring argument: among all optimal dual-server colorings, choose one with as few monochromatic edges as possible, where a monochromatic edge is an edge whose endpoints lie in the same color class, and then use a farthest such edge in a rooted tree to produce a contradiction.

Theorem 15. *If T is a nontrivial tree, then*

$$\gamma_{\mathrm{ds}}(T) = \gamma_{\mathrm{dsi}}(T).$$

Proof. The inequality $\gamma_{\mathrm{ds}}(T) \le \gamma_{\mathrm{dsi}}(T)$ is immediate. For the reverse inequality, among all γ_{ds}-sets of T and all associated red-blue partitions of such sets choose one, say $S = R \cup B$, that minimizes

$$m(R) + m(B),$$

where $m(X)$ is the number of edges in the induced subgraph $T[X]$. If $m(R) + m(B) = 0$, then S is already a DSI-dominating set.

Assume otherwise. Root T at a leaf r. Among all monochromatic edges in $T[R] \cup T[B]$, choose an edge uv as far from r as possible, where u is the parent of v. By interchanging colors if necessary, assume that u and v are red.

If v is a leaf, recolor v blue. This preserves the dual-server condition and decreases $m(R)+m(B)$, a contradiction. Hence, v has children. No child of v is red, by the choice of uv. For each child x of v, consider the subtree rooted at x. Recolor v blue and swap red and blue throughout each such subtree. The dual-server condition is preserved, the edge uv is no longer monochromatic, and no new monochromatic edge is created below v. This again contradicts the choice of the partition. Therefore, $m(R) + m(B) = 0$, and some γ_{ds}-set is a DSI-dominating set. Thus, $\gamma_{\mathrm{dsi}}(T) \le \gamma_{\mathrm{ds}}(T)$ and equality follows. □

Equality between the dual-server domination number and the 2-domination number of trees was proved in [7], yielding the following corollary of Theorem 15.

Corollary 16. *If T is a nontrivial tree, then*

$$\gamma_2(T) = \gamma_{\mathrm{ds}}(T) = \gamma_{\mathrm{dsi}}(T).$$

Hence, bounds on the dual-server domination number can be read as bounds for the dual-server independent domination in trees. We record one such bound, motivated by a path-covering bound of DeLaViña, Larson, Pepper, and Waller for 2-domination [16]. A *path covering* of a graph G is a collection of vertex-disjoint paths whose vertex sets partition $V(G)$. The *path covering number* $\rho(G)$ is the minimum number of paths in a path covering of G. We write $\nu(G)$ for the *matching number* of G (the maximum number of independent edges in G).

Theorem 17. *If G is a connected nontrivial graph, then*

$$\gamma_{\mathrm{ds}}(G) \leq \rho(G) + \nu(G).$$

Proof. Let $\mathcal{P} = \{P_1, \ldots, P_\rho\}$ be a minimum path covering of G, where

$$P_i = v_1^i v_2^i \cdots v_{n_i}^i.$$

For each i, let

$$D_i = \{v_j^i : j \text{ is odd}\} \cup \{v_{n_i}^i\}.$$

Thus, $|D_i| \leq \lfloor n_i/2 \rfloor + 1$. Color the vertices of D_i, in their natural order along P_i, alternately red and blue. Since G is connected and nontrivial, some path in $\mathcal{P}$ has order at least two, and hence, the two color classes may be chosen to be nonempty.

Let $S = \bigcup_{i=1}^{\rho} D_i$. If a vertex v_j^i is not in S, then j is even and $j < n_i$. Its two path-neighbors v_{j-1}^i and v_{j+1}^i both belong to D_i, and they receive different colors. Hence, every vertex outside S has a red neighbor and a blue neighbor, so S is a DS-dominating set.

For each path P_i, the edges

$$v_1^i v_2^i,\ v_3^i v_4^i,\ \ldots$$

form a matching of size $\lfloor n_i/2 \rfloor$. Since the paths in $\mathcal{P}$ are vertex-disjoint, the union of these matchings is a matching in G. Therefore,

$$|S| = \sum_{i=1}^{\rho} |D_i| \leq \sum_{i=1}^{\rho} \left(\left\lfloor \frac{n_i}{2} \right\rfloor + 1 \right) \leq \nu(G) + \rho(G).$$

It follows that $\gamma_{\mathrm{ds}}(G) \leq \rho(G) + \nu(G)$. ☐

Corollary 18. *If T is a nontrivial tree, then*

$$\gamma_{\mathrm{dsi}}(T) \leq \rho(T) + \nu(T).$$

The IDS version is more delicate in general, and is closely related to independent 2-domination. Since every IDS-dominating set is an independent 2-dominating set, any characterization of graphs with IDS-dominating sets must refine the known theory of independent 2-domination by adding the requirement that the independent 2-dominating set admit a red-blue partition that serves every external vertex in both colors. For trees, however, this extra coloring condition creates no additional obstruction. Indeed, if S is an independent 2-dominating set in a tree T, then for each vertex $u \in V(T) \setminus S$, choose two neighbors of u in S and join them by an auxiliary edge. The resulting auxiliary graph on S is a forest, for an auxiliary cycle would give a cycle in T. Thus, the auxiliary graph is bipartite, and a bipartition of it gives a red-blue partition of S in which every vertex outside S sees both colors. Consequently, the trees admitting IDS-dominating sets are precisely the trees admitting independent 2-dominating sets, which are characterized by Haynes, Hedetniemi, Henning, and Slater [25].

Problem 19. Characterize the graphs that admit an IDS-dominating set.

5 Isolation and General Bounds

The isolation number provides a tight lower bound on the dual-server independent domination number. An *isolating set* of G is a set $X \subseteq V(G)$ such that $G - N[X]$ is empty. The minimum cardinality of an isolating set is the *isolation number* $\iota(G)$. The isolation number was introduced by Caro and Hansberg in their work on partial domination [9]. It is also known as the vertex-edge domination number as first defined by Peters in [33]; see, for example, recent work of Goddard and Henning on upper bounds for this parameter under minimum-degree assumptions [21]. The relevance of this parameter is immediate from the definition of a DSI-partition: if $S = R \cup B$ is such a partition, then each of R and B must individually destroy all remaining edges after its closed neighborhood is removed.

This lower bound was generated by `Theo-Conjecture` as a conjecture; the equality characterization is proved here as part of the theorem.

Theorem 20. *For every nontrivial graph G,*

$$\gamma_{\mathrm{dsi}}(G) \geq 2\iota(G).$$

Moreover, equality holds if and only if the following conditions hold:

(i) there exist two disjoint sets $R, B \subseteq V(G)$;

(ii) R and B are independent sets;

(iii) $|R| = |B| = \iota(G)$; and

(iv) every vertex in $V(G) \setminus (R \cup B)$ has at least one neighbor in R and at least one neighbor in B.

Equivalently, equality holds if and only if G has a minimum DSI-dominating set whose associated red and blue color classes are both minimum isolating sets.

Proof. Let $S = R \cup B$ be a γ_{dsi}-set of G with associated DSI-partition. We first show that R is an isolating set. If an edge xy remained in $G - N[R]$, then neither x nor y is in R or adjacent to a vertex of R. Since B is independent, at least one of x and y is outside S. That vertex has no red neighbor, contradicting that S is a DSI-dominating set. Thus, $G - N[R]$ is empty, and so, $|R| \geq \iota(G)$. The same argument shows that B is an isolating set, and hence, $|B| \geq \iota(G)$. Therefore,

$$\gamma_{\mathrm{dsi}}(G) = |S| = |R| + |B| \geq 2\iota(G).$$

Suppose now that equality holds. By the preceding paragraph, R and B are isolating sets. Since

$$|R| + |B| = |S| = \gamma_{\mathrm{dsi}}(G) = 2\iota(G),$$

we have $|R| = |B| = \iota(G)$. The independence of R and B, and the dual-server domination of every vertex outside $R \cup B$, are part of the definition of a DSI-partition. Thus, conditions (i)–(iv) hold.

Conversely, if R and B are disjoint isolation sets of minimum cardinality, then $S = R \cup B$, with the associated partition, is a DSI-dominating set. Therefore,

$$\gamma_{\mathrm{dsi}}(G) \leq |S| = |R \cup B| = |R| + |B| = 2\iota(G).$$

Together with the lower bound already proved, this gives equality. $\square$

Remark 21. The equality statement in Theorem 20 can be viewed as a three-coloring criterion: equality holds precisely when $V(G)$ admits a red-blue-white coloring in which the red and blue classes are disjoint independent $\iota(G)$-sets and every white vertex sees both colors.

Example 22. Several familiar families satisfy the equality criterion in Remark 21. Using the standard formulas [9],

$$\iota(P_n) = \left\lceil \frac{n-1}{4} \right\rceil \quad \text{and} \quad \iota(C_n) = \left\lceil \frac{n}{4} \right\rceil,$$

together with Propositions 13 and 14, we see that every cycle C_n, $n \geq 3$, satisfies

$$\gamma_{\mathrm{dsi}}(C_n) = 2\iota(C_n),$$

while a path P_n, $n \geq 2$, satisfies

$$\gamma_{\mathrm{dsi}}(P_n) = 2\iota(P_n)$$

if and only if $n \equiv 2, 3 \pmod 4$. The criterion also captures a broad family with isolation number one: if G has two vertices x and y such that every vertex in $V(G) \setminus \{x, y\}$ is adjacent to both x and y, then $\iota(G) = 1$, and coloring x red and y blue gives $\gamma_{\mathrm{dsi}}(G) = 2$. This includes complete graphs, complete bipartite graphs $K_{2,s}$, double cones of the form $K_1 \vee 2C_k$, and more generally joins of the form $K_2 \vee H$ and $\overline{K_2} \vee H$.

Proposition 23. *For every positive integer k, there are connected graphs G with $\iota(G) = k$ and $\gamma_{\mathrm{dsi}}(G) = 2k$.*

Proof. For each $i \in [k]$, take four vertices r_i, b_i, u_i, v_i and add the five edges

$$u_iv_i, \quad r_iu_i, \quad r_iv_i, \quad b_iu_i, \quad b_iv_i.$$

Thus, each set $\{r_i, b_i, u_i, v_i\}$ induces a diamond $K_4 - e$, with missing edge r_ib_i. Let

$$R = \{r_i : i \in [k]\} \quad \text{and} \quad B = \{b_i : i \in [k]\}.$$

Now add any collection of edges with one endpoint in R and one endpoint in B, chosen so that the resulting graph is connected; for instance, one may add the edges b_ir_{i+1} for $1 \leq i < k$. Denote the resulting graph by G. The case $k = 3$ is illustrated in Figure 1.

The sets R and B are independent, since all added edges go between R and B. Moreover, every vertex u_i and v_i is adjacent to both r_i and b_i. Hence, $R \cup B$, with color classes R and B, is a DSI-dominating set of G. Also, R is an isolating set: all vertices u_i and v_i lie in $N[R]$, all vertices of R lie in $N[R]$, and the only possible remaining vertices are in B, which is independent. Thus, $\iota(G) \leq k$. Similarly, B is an isolating set.

For the reverse inequality, observe that the edge u_iv_i can be deleted by an isolating set X only if X contains at least one vertex from $\{r_i, b_i, u_i, v_i\}$. These four-vertex sets are pairwise disjoint as i varies, and so every isolating set has cardinality at least k. Consequently, $\iota(G) = k$. The sets R and B therefore satisfy the equality of Theorem 20, and hence, $\gamma_{\mathrm{dsi}}(G) = 2k$. □

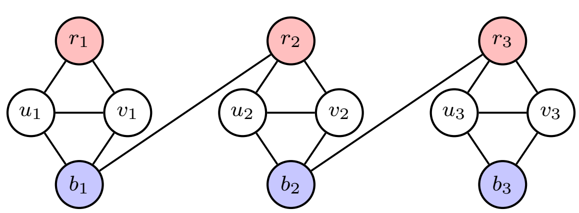

Figure 1: A connected graph G with $\iota(G) = 3$ and $\gamma_{\mathrm{dsi}}(G) = 6$.

We next give a lower bound on the dual-server independent domination number of a graph G in terms of its diameter, denoted $\mathrm{diam}(G)$. Let $d_G(u, v)$ denote the distance between vertices u and v in G.

This inequality was generated by `Theo-Conjecture` as a conjecture.

Theorem 24. *If G is a nontrivial connected graph, then*

$$\gamma_{\mathrm{dsi}}(G) \geq \left\lceil \frac{\mathrm{diam}(G)}{2} \right\rceil + 1.$$

Moreover, the bound is sharp for every possible value of the diameter.

Proof. Let $d = \mathrm{diam}(G)$, and let

$$P = v_0 v_1 \cdots v_d$$

be a diametral path in G. Let $S = R \cup B$ be a DSI-dominating set of G, and let $k = |S|$ and $t = |S \cap V(P)|$. We count the pairs (s, v_i), where $s \in S$, $0 \leq i \leq d$, and $s \in N[v_i]$.

For a fixed vertex $s \in S$, there are at most three indices i for which $s \in N[v_i]$. Indeed, if $s \in N[v_i] \cap N[v_j]$, then $d_G(v_i, v_j) \leq 2$. Since P is a geodesic path, the subpath of P from v_i to v_j is a shortest v_i-v_j-path, and hence, $|i - j| \leq 2$. Thus, the indices i with $s \in N[v_i]$ are contained in three consecutive positions. Therefore, the total number of counted pairs is at most $3k$.

On the other hand, each vertex of $S \cap V(P)$ contributes at least one pair, namely the pair in which it counts itself. Each vertex of $V(P) \setminus S$ has a red neighbor and a blue neighbor in S, and these two neighbors are distinct. Hence, each vertex of $V(P) \setminus S$ contributes at least two pairs. It follows that

$$3k \geq t + 2(d + 1 - t) = 2d + 2 - t.$$

Since $t \leq k$, we obtain

$$3k \geq 2d + 2 - k,$$

and so $4k \geq 2d + 2$. Consequently, $k \geq (d + 1)/2$.

If d is even, then since k is an integer, this gives

$$k \geq \left\lceil \frac{d}{2} \right\rceil + 1.$$

Assume now that $d = 2q + 1$ is odd. The preceding inequality gives $k \geq q + 1$. Suppose, for a contradiction, that $k = q + 1$. Then equality holds throughout the preceding counting argument. In particular $t = k$, so every vertex of S lies on P, and every selected vertex must be counted by exactly three vertices of P. Since all selected vertices lie on P, neither endpoint v_0 nor v_d can lie outside S, for otherwise, it would have at most one selected neighbor on P and no selected neighbor outside P. Thus, $\{v_0, v_d\} \subseteq S$. But an endpoint of a geodesic path is contained in the closed neighborhoods of at most two vertices of P, contradicting the requirement that every selected vertex be counted exactly three times. Therefore, $k \geq q + 2$, which is precisely

$$k \geq \left\lceil \frac{d}{2} \right\rceil + 1.$$

Since S was arbitrary, the desired lower bound follows.

To see that the bound is sharp, take $G = P_n$ with $n \geq 2$. By Proposition 13,

$$\gamma_{\text{dsi}}(P_n) = \left\lceil \frac{n+1}{2} \right\rceil.$$

Since $\text{diam}(P_n) = n - 1$, this is exactly

$$\left\lceil \frac{\text{diam}(P_n)}{2} \right\rceil + 1.$$

Thus, equality holds for paths, and hence, for every positive value of the diameter. $\square$

We note that there is no forbidden subgraph characterization of graphs attaining the bound of Theorem 24. To show this, we construct a family of extremal graphs admitting any arbitrary subgraph. For $d \geq 2$, let $\mathcal{F}_d$ be the family of graphs that can be obtained from a path $P = v_0 v_1 \cdots v_d$ with a given γ_{dsi}-set $S = R \cup B$ and associated DSI-partition. For each vertex $v_i \notin S$, replace v_i by an arbitrary nonempty graph H_i, joining every vertex of H_i to v_{i-1} and v_{i+1}, and adding no other edges incident with H_i. Keep the selected vertices of P unchanged. For an example of a graph $G \in \mathcal{F}_6$, see Figure 2, where $R = \{v_0, v_4\}$, $B = \{v_2, v_6\}$, and $S = R \cup B$. The selected vertices v_0, v_2, v_4, v_6 retain the optimal red-blue coloring of the path, while the unselected vertices v_1, v_3, v_5 have been replaced by nonempty white graphs $H_1 = K_2, H_3 = \overline{K}_2$, and $H_5 = K_1$. Every white vertex sees both adjacent colors, and the structure preserves the diameter of the path.

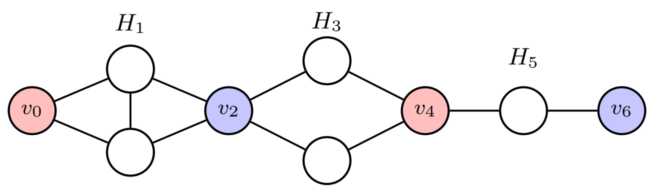

Figure 2: A graph $G \in \mathcal{F}_6$ with $\gamma_{\text{dsi}}(G) = 4$ and $\text{diam}(G) = d = 6$.

Proposition 25. *If $G \in \mathcal{F}_d$, then*

$$\text{diam}(G) = d \quad \textit{and} \quad \gamma_{\text{dsi}}(G) = \left\lceil \frac{d}{2} \right\rceil + 1.$$

Proof. For $d \geq 2$, let $G \in \mathcal{F}_d$ be obtained from the path $P = v_0 v_1 \cdots v_d$ with a given γ_{dsi}-set $S = R \cup B$ and associated DSI-partition. By the construction of G, we deduce that $S = R \cup B$ is a a DSI-dominating set of G. Consequently,

$$\gamma_{\text{dsi}}(G) \leq |S| = \gamma_{\text{dsi}}(P_{d+1}) = \left\lceil \frac{d+2}{2} \right\rceil = \left\lceil \frac{d}{2} \right\rceil + 1.$$

It remains to check that $\text{diam}(G) = d = \text{diam}(P)$. By Observation 1, the vertices v_0 and v_d are in S, and hence are not replaced when forming G. By the construction of G, we deduce that $d_G(v_0, v_d) = d_P(v_0, v_d) = d$, implying that $\text{diam}(G) \geq d$.

Conversely, we observe that $\text{diam}(G) \leq d$ as follows. For a vertex $v_i \in V(P) \setminus S$, we let u_i denote an arbitrary vertex in the replacement graph H_i. For any $i, j, k \in [d]$, we deduce, from the construction of G, that $d_G(v_i, u_j) = d_P(v_i, v_j) \leq d$ and $d_G(u_j, u_k) = d_P(v_j, v_k) \leq d$, where $v_i \in S$, $v_j, v_k \in V(P) \setminus S$, $u_j \in V(H_j)$, and $u_k \in V(H_k)$. Moreover, the construction implies that

$d_G(v_i, v_j) = d_P(v_i, v_j) \leq d$ for any two vertices v_i and v_j in S. Finally, for a vertex $v_i \in V(P) \setminus S$, any two vertices in the replacement graph H_i are at distance at most $2 \leq d$ in G, since both are adjacent to v_{i-1} and v_{i+1}.

Hence, $\operatorname{diam}(G) \leq d$, and so $\operatorname{diam}(G) = d$. The lower bound in Theorem 24 now forces equality for $\gamma_{\mathrm{dsi}}(G)$. $\square$

Corollary 26. *There is no forbidden subgraph characterization of the graphs attaining the lower bound of Theorem 24.*

The *decycling number* $\operatorname{dec}(G)$ is the minimum cardinality of a set whose deletion leaves a forest, and the *forest number* $\operatorname{forest}(G)$ is the maximum order of an induced forest in G.

Proposition 27. *For every graph G of order n,*

$$\operatorname{forest}(G) + \operatorname{dec}(G) = n.$$

Proof. Let P be the property of being a forest. This property is hereditary in the sense of Hedetniemi [28]: every subgraph of a forest is a forest. In the notation of [28], $\beta_0(P)$ is the maximum number of vertices in a set that induces a graph with property P, and $\alpha_0(P)$ is the minimum number of vertices in a set meeting every subgraph that does not have property P. For P equal to being a forest, these two quantities are precisely $\operatorname{forest}(G)$ and $\operatorname{dec}(G)$, respectively: a vertex set meets every nonforest subgraph if and only if its deletion leaves a forest. Hedetniemi's Theorem [28, Theorem 2] gives $\alpha_0(P) + \beta_0(P) = n$, and the result follows. $\square$

The next proposition verifies, under the additional hypothesis that G has an IDS-dominating set, the `Theo-Conjecture` conjecture $\gamma_{\mathrm{dsi}}(G) \leq n - \operatorname{dec}(G)$.

Proposition 28. *If G is a graph of order n and G has an IDS-dominating set, then*

$$\gamma_{\mathrm{dsi}}(G) \leq i_{\mathrm{ds}}(G) \leq \operatorname{forest}(G) = n - \operatorname{dec}(G).$$

Proof. Let S be an i_{ds}-set of G. Since S is independent, $G[S]$ is an empty forest, and so $|S| \leq \operatorname{forest}(G)$. Thus, $i_{\mathrm{ds}}(G) \leq \operatorname{forest}(G)$. The equality $\operatorname{forest}(G) = n - \operatorname{dec}(G)$ follows from Proposition 27, and $\gamma_{\mathrm{dsi}}(G) \leq i_{\mathrm{ds}}(G)$ follows from Observation 7. $\square$

For DSI-domination, the corresponding hereditary property is induced bipartiteness. Let $\operatorname{bipartite}(G)$ be the *bipartite number* of G, the maximum order of an induced bipartite subgraph of G, and let $\operatorname{bipartite-cover}(G)$ be the *bipartite covering number* of G, the minimum cardinality of a set whose deletion leaves a bipartite graph.

Proposition 29. *For every graph G of order n,*

$$\operatorname{bipartite-cover}(G) + \operatorname{bipartite}(G) = n.$$

Proof. This is the instance of Hedetniemi's Theorem [28, Theorem 2] for the hereditary property of being bipartite. $\square$

Proposition 30. *For every nontrivial graph G,*

$$\gamma_{\rm dsi}(G) \le \text{bipartite}(G).$$

Furthermore, if G has an IDS-set, then

$$\gamma_{\rm dsi}(G) \le i_{\rm ds}(G) \le \alpha(G) \le \text{bipartite}(G).$$

Proof. Let S be a $\gamma_{\rm dsi}$-set of G, and let $S = R \cup B$ be a DSI-partition. Since R and B are independent, $G[S]$ is bipartite with bipartition $\{R, B\}$. Hence, $|S| \le \text{bipartite}(G)$, and therefore, $\gamma_{\rm dsi}(G) \le \text{bipartite}(G)$.

If G has an $i_{\rm ds}$-set S, then since S is independent, $2 \le i_{\rm ds}(G) = |S| \le \alpha(G)$. Every independent set of cardinality at least 2 induces a bipartite, so $\alpha(G) \le \text{bipartite}(G)$. □

6 Computation and Complexity

We next present both a concrete optimization model for computing the dual-server independent domination number and the corresponding hardness result. The optimization model is also useful for reproducibility and for automated conjecturing. In the `Theo-Conjecture` experiments, the *Add Definition* tool was used to add $\gamma_{\rm dsi}$ to the First Principles workspace parameter database. In this workflow, the user supplied the mathematical definition, and the system drafted a user-defined optimization model that was checked on small examples before being used in larger searches. Once saved, the invariant could be evaluated on graph families and example databases in the same way as the built-in parameters, so that conjecture searches could compare $\gamma_{\rm dsi}$ with domination, distance, independence, hereditary-subgraph, and other structural quantities. The binary integer programming model below is the mathematical form of the entry used for these computations.

The manuscript source, scripts, CSV files, LP files, CBC logs, solution files, fixed seeds, and metadata for the computational experiments are available in the public artifact repository `https://github.com/RandyRDavila/dsi-domination-artifacts`. Paths cited below are relative to the root of that repository.

Let G be a nontrivial graph. For each vertex $v \in V(G)$, introduce binary variables r_v and b_v, where $r_v = 1$ means that v is selected in the red class and $b_v = 1$ means that v is selected in the blue class. Consider the following binary integer program:

$$\begin{aligned}
\text{minimize} \quad & \sum_{v\in V(G)} (r_v + b_v) \\
\text{subject to} \quad & \sum_{v\in V(G)} r_v \geq 1, \\
& \sum_{v\in V(G)} b_v \geq 1, \\
& r_v + b_v \leq 1 \qquad \text{for every } v \in V(G), \\
& r_u + r_v \leq 1 \qquad \text{for every } uv \in E(G), \\
& b_u + b_v \leq 1 \qquad \text{for every } uv \in E(G), \\
& r_v + b_v + \sum_{u\in N(v)} r_u \geq 1 \qquad \text{for every } v \in V(G), \\
& r_v + b_v + \sum_{u\in N(v)} b_u \geq 1 \qquad \text{for every } v \in V(G), \\
& r_v, b_v \in \{0,1\} \qquad \text{for every } v \in V(G).
\end{aligned} \tag{IP}$$

The last two constraint families are the two-color domination constraints. They say that each vertex v is either selected, meaning $r_v + b_v = 1$, or has a red neighbor; and, independently, that v is either selected or has a blue neighbor. Thus, if $v \notin R\cup B$, then $r_v = b_v = 0$, and both a red neighbor and a blue neighbor are required. If $v \in R \cup B$, then no domination requirement is imposed on v itself.

Proposition 31. *For every nontrivial graph G, the optimum value of* (IP) *is* $\gamma_{\mathrm{dsi}}(G)$.

Proof. Given a feasible solution of (IP), let

$$R = \{v \in V(G) : r_v = 1\} \qquad \text{and} \qquad B = \{v \in V(G) : b_v = 1\}.$$

The first two constraints require R and B to be nonempty, and the third requires them to be disjoint. The edge constraints $r_u + r_v \leq 1$ and $b_u + b_v \leq 1$ require R and B to be independent. Finally, if $v \notin R\cup B$, then $r_v = b_v = 0$, and the two neighborhood constraints require v to have a neighbor in R and a neighbor in B. Thus, $R \cup B$, with partition $\{R, B\}$, is a DSI-dominating set, and its cardinality is the value of the objective function.

Conversely, if $S = R \cup B$ is a DSI-dominating set with associated DSI-partition, set $r_v = 1$ for $v \in R$, set $b_v = 1$ for $v \in B$, and set all other variables equal to 0. The DSI conditions give every constraint of (IP), and the objective value is $|S|$. Therefore, the minimum possible objective value is exactly $\gamma_{\mathrm{dsi}}(G)$. □

We now turn to the decision version and show that, despite this compact model, computing $\gamma_{\mathrm{dsi}}(G)$ is computationally NP-complete. Recall that a bipartite graph is *chordal bipartite* if every cycle of length at least six has a chord. The relevant decision problem is the following.

DSI-Domination.

Instance: A nontrivial graph G of order n and a positive integer $k \leq n$.

Question: Does G have a DSI-dominating set of cardinality at most k?

Theorem 32. DSI-Domination *is NP-complete, even when restricted to chordal bipartite graphs.*

Proof. The problem belongs to $\mathcal{NP}$, since a certificate consists of two sets $R, B \subseteq V(G)$. In polynomial time one verifies that R and B are nonempty and disjoint, that both R and B are independent, that $|R \cup B| \leq k$, and that every vertex outside $R \cup B$ has a neighbor in each of R and B.

For hardness, we reduce from Dominating Set restricted to chordal bipartite graphs, which is NP-complete by a theorem of Müller and Brandstädt [32]. Let H be a chordal bipartite graph, let k be a positive integer with $k \leq |V(H)|$, and write $n = |V(H)|$. Construct a graph G from H by adding, for each $v \in V(H)$, a new leaf x_v adjacent only to v. Let

$$L = \{x_v : v \in V(H)\} \qquad \text{and} \qquad K = n + k.$$

The construction is polynomial. Since adding leaves creates no new cycles, G is chordal bipartite whenever H is chordal bipartite.

Suppose first that H has a dominating set D with $|D| \leq k$. Fix a bipartition (A, B) of H. We claim that $S = L \cup D$ is a DSI-dominating set of G. Color the vertices of $D \cap A$ red and the vertices of $D \cap B$ blue. For each $v \in D$, color x_v opposite to the color of v. For each $v \notin D$, color x_v red if $v \in A$ and blue if $v \in B$. This is a proper red-blue coloring of $G[S]$, so both color classes are independent. Now let $v \in V(H) \setminus D$. The leaf x_v has the color corresponding to the part of H containing v, and since D dominates H, the vertex v has a neighbor $u \in D$ in the other part, whose color is opposite to the color of x_v. Thus, every vertex outside S sees both colors, and $|S| = n + |D| \leq K$.

Conversely, suppose that G has a DSI-dominating set $S = R \cup B$ with $|S| \leq K$. Every leaf x_v belongs to S, for otherwise x_v would have at most one selected neighbor and could not see both colors. Let $D = S \cap V(H)$. Since $L \subseteq S$, we have

$$|D| \leq |S| - |L| \leq K - n = k.$$

If $v \in V(H) \setminus D$, then $v \notin S$. Its selected neighbors in G consist of x_v and the vertices in $D \cap N_H(v)$. Since v must see both color classes, it follows that $D \cap N_H(v) \neq \emptyset$. Thus, D is a dominating set of H of cardinality at most k. This proves the equivalence of the two decision instances. $\square$

Proposition 33. *For every fixed integer $t \geq 0$, the value $\gamma_{\text{dsi}}(G)$ can be computed in linear time on the class of nontrivial graphs satisfying treewidth* $\text{tw}(G) \leq t$. *Consequently,* DSI-Domination *is polynomial-time solvable on every graph class of bounded treewidth.*

Proof. We use the standard monadic second-order optimization framework for graphs of bounded treewidth [10, 11]. Let $x \sim y$ denote adjacency. For vertex-set variables R and B, consider the following monadic second-order formula:

$$\begin{aligned}\Phi(R, B) \equiv\ & (\exists r\, r \in R) \wedge (\exists b\, b \in B) \\ & \wedge \forall x\, \neg(x \in R \wedge x \in B) \\ & \wedge \forall x \forall y\, \big((x \in R \wedge y \in R \wedge x \sim y) \to x = y\big) \\ & \wedge \forall x \forall y\, \big((x \in B \wedge y \in B \wedge x \sim y) \to x = y\big) \\ & \wedge \forall x\, \big((x \notin R \wedge x \notin B) \to (\exists y\, (y \in R \wedge x \sim y) \wedge \exists z\, (z \in B \wedge x \sim z))\big).\end{aligned}$$

The first line requires both color classes to be nonempty, the second requires them to be disjoint, the next two require each color class to be independent, and the final line requires every unselected vertex to have a neighbor in each color class. Hence, a pair (R, B) satisfies Φ precisely when $R \cup B$, with its prescribed partition, is a DSI-dominating set. Therefore,

$$\gamma_{\mathrm{dsi}}(G) = \min\{|R| + |B| : G \models \Phi(R, B)\}.$$

Since Φ is fixed, the Monadic Second-Order Optimization Theorem gives a linear-time algorithm, for each fixed t, to compute this minimum from a tree-decomposition of width at most t. Bodlaender's algorithm finds such a tree-decomposition in linear time for fixed t, or correctly reports that the treewidth exceeds t [6]. Combining these two algorithms proves the claim. □

7 `Theo-Conjecture` Open Problems

We conclude this paper with a sampling of the conjectures generated by `Theo-Conjecture` after the parameter γ_{dsi} was added to the First Principles automated conjecturing system. The conjectures below resisted both human counterexample searches and counterexample attempts made through the *Co-Pilot* chat interface with the system. They represent our favorite surviving conjectures from this process; several other system-generated conjectures were found to be false, either by the authors or by the AI system itself.

We use $\overline{d}(G)$ for the *average distance* of a connected graph G. The *domination core* of G, denoted $\mathrm{Core}_\gamma(G)$, is the set of vertices that belong to every γ-set of G. A *paired dominating set* of G is a dominating set D such that $G[D]$ contains a perfect matching; its minimum cardinality is the *paired domination number* $\gamma_{\mathrm{pr}}(G)$ [20]. The conjectures are ordered as follows: first a mixed degree and paired domination inequality, then two inequalities of the form $\gamma_{\mathrm{dsi}}(G) + \cdots \leq n$, followed by two lower bounds, two upper bounds, and finally an equality conjecture.

Conjecture 34 (`Theo-Conjecture`)**.** *If G is a nontrivial connected graph, then*

$$\delta(G)\gamma_{\mathrm{pr}}(G) \leq \Delta(G)\gamma_{\mathrm{dsi}}(G).$$

Conjecture 35 (`Theo-Conjecture`)**.** *If G is a nontrivial graph of order n, then*

$$\gamma_{\mathrm{dsi}}(G) + \mathrm{dec}(G) \leq n.$$

Conjecture 36 (`Theo-Conjecture`)**.** *If G is a nontrivial connected graph, then*

$$\gamma_{\mathrm{dsi}}(G) + |\,\mathrm{Core}_\gamma(G)| \leq n.$$

Conjecture 37 (`Theo-Conjecture`)**.** *If G is a nontrivial connected claw-free graph, then*

$$\gamma_{\mathrm{dsi}}(G) \geq \left\lceil \frac{3\overline{d}(G)}{2} \right\rceil.$$

Conjecture 38 (`Theo-Conjecture`)**.** *If G is a nontrivial connected triangle-free graph, then*

$$\gamma_{\mathrm{dsi}}(G) \geq \mathrm{dec}(G) + 1.$$

Conjecture 39 (`Theo-Conjecture`)**.** *If G is a nontrivial connected claw-free graph, then*

$$\gamma_{\mathrm{dsi}}(G) \leq \left\lfloor \frac{3\,\mathrm{bipartite}(G)}{4} \right\rfloor + 1.$$

Conjecture 40 (`Theo-Conjecture`). *If G is a nontrivial connected cubic graph, then*

$$\gamma_{\mathrm{dsi}}(G) \leq 2\gamma(G).$$

Conjecture 41 (`Theo-Conjecture`). *If G is a nontrivial connected bipartite cubic graph, then*

$$\gamma_{\mathrm{dsi}}(G) = \gamma_{\mathrm{ds}}(G).$$

A Baseline Ensemble Illustrations

This appendix records two small baseline studies connected with the ensemble-learning interpretation mentioned in the introduction. They are included for reproducibility and motivation only. The purpose is not to propose a new ensemble-learning algorithm, nor to claim that DSI-domination is a competitive classifier-selection method. Instead, the studies illustrate how the DSI constraint can be implemented as an exact graph-theoretic diversity-and-coverage rule on a learner-dependence graph. The first study uses synthetic data with known latent dependence families. The second uses standard classification datasets and adds a simple accuracy-aware refinement after the minimum DSI cardinality has been fixed.

A.1 Synthetic Baseline

In each trial, we generated 48 binary base predictors arranged in six latent dependence families of eight predictors each. For each observation, all predictors in the same family were affected by a shared error source, and each predictor was also affected by an independent error source. The learner graph was constructed from validation data by joining two predictors when the phi-correlation of their validation error indicators was at least 0.15. Thus, adjacency represents a strong tendency to err together. We then solved the binary integer program (IP) for a DSI-dominating set, used the selected predictors in a majority vote on an independent test set, and compared this vote with 200 uniformly random same-size subsets of predictors. The baseline used 30 fixed random seeds, 1000 validation observations, and 3000 test observations per trial.

The raw data, fixed seeds, LP files, CBC logs, solution files, and metadata are available in the artifact repository at `https://github.com/RandyRDavila/dsi-domination-artifacts/tree/main/computational_data/ensemble`. The repository-relative script `computational_data/ensemble_dsi_experiment.py` regenerates the baseline using only the Python standard library and the command-line CBC solver.

Quantity	Value
Trials	30
Base predictors	48
Latent dependence families	6
Random same-size ensembles per trial	200
Trials solved optimally by CBC	30
Mean γ_{dsi} of learner graph	12.000
Mean DSI majority-vote accuracy	0.953911
Mean random same-size majority-vote accuracy	0.933030
Mean accuracy difference	0.020881
Trials where DSI exceeded the random mean	30
Mean DSI accuracy percentile among random subsets	0.951667
Mean DSI average error correlation	0.047216
Mean random average error correlation	0.078203
Mean correlation difference	-0.030986

Table 1: Aggregate results for the synthetic ensemble baseline. The random accuracy and random error-correlation values are trialwise averages over 200 random same-size subsets.

In this synthetic model, the validation threshold recovered the latent dependence families, and a minimum DSI-dominating set selected two representatives from each family, one in each color class. The DSI selection therefore had lower average pairwise error correlation than a random same-size selection, while preserving coverage of the learner pool through the domination condition. This is the only point of the synthetic baseline: it verifies, in a controlled setting, that the graph-theoretic rule behaves as the definition suggests.

A.2 Standard-Data Baseline

The second baseline uses four standard classification datasets: Iris [17], Palmer Penguins [26], Wine [1], and Wisconsin Diagnostic Breast Cancer [37]. In each trial, the data were split in a stratified 60/20/20 train-validation-test proportion. We trained 48 simple base learners using only the Python standard library: nearest-centroid classifiers, Gaussian naive Bayes classifiers, k-nearest-neighbor classifiers, and decision stumps, with both full and random feature subsets. Since diversity is useful only among competent predictors, we first retained the learners whose validation accuracy was within 0.05 of the best validation accuracy, keeping at least 16 candidates. The learner graph was then formed on this candidate pool by joining the top 25% of pairs by validation-set prediction agreement. A minimum DSI-dominating set was computed by the binary integer program from Section 6, and its majority vote was evaluated on the test set. We also computed an accuracy-aware lexicographic refinement. Namely, after first determining γ_{dsi} for the learner graph, we maximized, among DSI-dominating sets of this minimum cardinality, the number of validation examples on which the correct class received a strict plurality vote from the selected learners. Individual validation accuracy was used only as a small tie-breaker. We call this second-stage rule VM-DSI. It preserves the exact DSI size constraint while letting the second-stage optimization distinguish between minimum DSI sets that are equally good from the graph-theoretic point of view.

The raw data files, fixed seeds, trial-level CSV file, summary CSV file, LP files, CBC logs, solution files, and metadata are available in the artifact repository at `https://github.com/RandyRDavila/dsi-domination-artifacts/tree/main/computational_data/real_data`. The regeneration script is available at `computational_data/real_data_dsi_experiment.py` and uses only the Python standard library and the command-line CBC solver.

Dataset	Cand.	γ_{dsi}	DSI acc.	VM-DSI acc.	Top-k acc.	Rand. acc.	Full acc.	VM agr.	Rand. agr.
Iris	17.950	6.650	0.940	0.947	0.950	0.944	0.945	0.946	0.951
Penguins	16.000	8.800	0.964	0.965	0.992	0.985	0.989	0.808	0.862
WDBC	16.150	7.800	0.954	0.958	0.964	0.957	0.961	0.919	0.928
Wine	16.000	6.800	0.969	0.963	0.961	0.966	0.971	0.883	0.901

Table 2: Aggregate results for the standard-data ensemble baseline. Each row reports 20 stratified train-validation-test splits. "Cand." is the mean number of validation-competent candidate learners. VM-DSI is the validation-margin DSI refinement. The random accuracy and random agreement values are trialwise averages over 200 random same-size subsets of the candidate pool. Agreement is the mean pairwise test-set prediction agreement within the selected ensemble.

These data should be read cautiously. The minimum DSI rule is an exact graph-theoretic selection rule, not an accuracy-aware ensemble objective. The validation-margin refinement is more faithful to the ensemble-learning interpretation: it keeps the same DSI cardinality but uses validation data to choose among the minimum DSI sets. Even then, the goal is illustrative rather than competitive. On these small datasets, VM-DSI keeps test accuracy near the natural baselines while retaining lower pairwise agreement than random same-size selections. Thus, the standard-data baseline supports a modest interpretation: DSI-domination can serve as a mathematically clean diversity-and-coverage primitive inside a larger ensemble-selection pipeline, but further accuracy-aware variants would be needed before presenting it as a standalone machine learning method.

References

[1] S. Aeberhard and M. Forina, Wine [Dataset], UCI Machine Learning Repository, 1992, doi:10.24432/C5PC7J.

[2] R. B. Allan and R. Laskar, On domination and independent domination numbers of a graph, *Discrete Math.* **23** (1978), 73–76.

[3] M. Blidia, M. Chellali, and O. Favaron, Ratios of some domination parameters in graphs and claw-free graphs, in *Graph Theory*, Trends in Mathematics, Birkhäuser, Basel, 2006, 61–72.

[4] M. Blidia, M. Chellali, and O. Favaron, Independence and 2-domination in trees, *Australas. J. Combin.* **33** (2005), 317–327.

[5] M. Blidia, M. Chellali, and L. Volkmann, Bounds of the 2-domination number of graphs, *Utilitas Math.* **71** (2006), 209–216.

[6] H. L. Bodlaender, A linear-time algorithm for finding tree-decompositions of small treewidth, *SIAM J. Comput.* **25** (1996), 1305–1317.

[7] M. Chellali, T. W. Haynes, and S. T. Hedetniemi, Dual-server domination in graphs, *Discrete Appl. Math.* **387** (2026), 251–259.

[8] G. Brown, J. Wyatt, R. Harris, and X. Yao, Diversity creation methods: a survey and categorisation, *Information Fusion* **6** (2005), 5–20.

[9] Y. Caro and A. Hansberg, Partial domination – the isolation number of a graph, *Filomat* **31** (2017), 3925–3944.

[10] B. Courcelle, The monadic second-order logic of graphs. I. Recognizable sets of finite graphs, *Inform. and Comput.* **85** (1990), 12–75.

[11] B. Courcelle, J. A. Makowsky, and U. Rotics, Linear time solvable optimization problems on graphs of bounded clique-width, *Theory Comput. Syst.* **33** (2000), 125–150.

[12] R. Davila, Automated conjecturing with TxGraffiti, *Ann. Math. Artif. Intell.* (2026), doi:10.1007/s10472-026-10005-5.

[13] R. Davila, Graffiti3: Compact theory libraries for automated mathematical discovery, Research Square preprint, 2026, doi:10.21203/rs.3.rs-8493329/v1.

[14] T. G. Dietterich, Ensemble methods in machine learning, in *Multiple Classifier Systems*, Lecture Notes in Computer Science **1857**, Springer, Berlin, 2000, 1–15.

[15] A. Dafoe, E. Hughes, Y. Bachrach, T. Collins, K. R. McKee, J. Z. Leibo, K. Larson, and T. Graepel, Open problems in cooperative AI, arXiv:2012.08630, 2020.

[16] E. DeLaViña, C. E. Larson, R. Pepper, and B. Waller, Graffiti.pc on the 2-domination number of a graph, *Congr. Numer.* **203** (2010), 15–32.

[17] R. A. Fisher, Iris [Dataset], UCI Machine Learning Repository, 1936, doi:10.24432/C56C76.

[18] J. F. Fink and M. S. Jacobson, n-domination in graphs, in *Graph Theory with Applications to Algorithms and Computer Science*, Wiley, New York, 1985, 283–300.

[19] F. Harary and T. W. Haynes, Double domination in graphs, *Ars Combin.* **55** (2000), 201–213.

[20] T. W. Haynes and P. J. Slater, Paired-domination in graphs, *Networks* **32** (1998), 199–206.

[21] W. Goddard and M. A. Henning, On the isolation number of graphs with minimum degree four, arXiv:2508.21551, 2025.

[22] T. W. Haynes, S. T. Hedetniemi, and M. A. Henning (eds.), *Topics in Domination in Graphs*, Developments in Mathematics **64**, Springer, Cham, 2020.

[23] T. W. Haynes, S. T. Hedetniemi, and M. A. Henning (eds.), *Structures of Domination in Graphs*, Developments in Mathematics **66**, Springer, Cham, 2021.

[24] T. W. Haynes, S. T. Hedetniemi, and M. A. Henning, *Domination in Graphs: Core Concepts*, Springer Monographs in Mathematics, Springer, Cham, 2023.

[25] T. W. Haynes, S. T. Hedetniemi, M. A. Henning, and P. J. Slater, H-forming sets in graphs, *Discrete Math.* **262** (2003), 159–169.

[26] A. M. Horst, A. P. Hill, and K. B. Gorman, *palmerpenguins: Palmer Archipelago (Antarctica) Penguin Data*, R package version 0.1.0, 2020, doi:10.5281/zenodo.3960218.

[27] S. M. Hedetniemi, S. T. Hedetniemi, R. C. Laskar, L. Markus, and P. J. Slater, Disjoint dominating sets in graphs, *Proc. Internat. Conf. Discrete Math., ICDM 2006*, Ramanujan Math. Soc. Lect. Notes Ser. **7** (2008), 87–100.

[28] S. T. Hedetniemi, Hereditary properties of graphs, *J. Combin. Theory Ser. B* **14** (1973), 94–99.

[29] G. Irving, P. Christiano, and D. Amodei, AI safety via debate, arXiv:1805.00899, 2018.

[30] L. I. Kuncheva, *Combining Pattern Classifiers: Methods and Algorithms*, John Wiley & Sons, Hoboken, NJ, 2004.

[31] Y. Liu and X. Yao, Ensemble learning via negative correlation, *Neural Networks* **12** (1999), 1399–1404.

[32] H. Müller and A. Brandstädt, The NP-completeness of Steiner tree and dominating set for chordal bipartite graphs, *Theoret. Comput. Sci.* **53** (1987), 257–265.

[33] K. W. Peters, *Theoretical and Algorithmic Results on Domination and Connectivity*, Ph.D. thesis, Clemson University, 1986.

[34] Y. Shoham and K. Leyton-Brown, *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*, Cambridge University Press, Cambridge, 2008.

[35] D. P. Sumner, Critical concepts in domination, *Discrete Math.* **86** (1990), 33–46.

[36] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed., John Wiley & Sons, Chichester, 2009.

[37] W. Wolberg, O. Mangasarian, N. Street, and W. Street, Breast Cancer Wisconsin (Diagnostic) [Dataset], UCI Machine Learning Repository, 1993, doi:10.24432/C5DW2B.